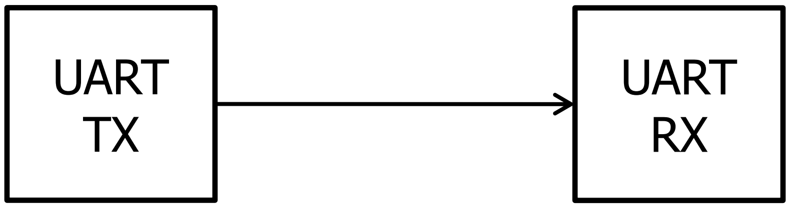
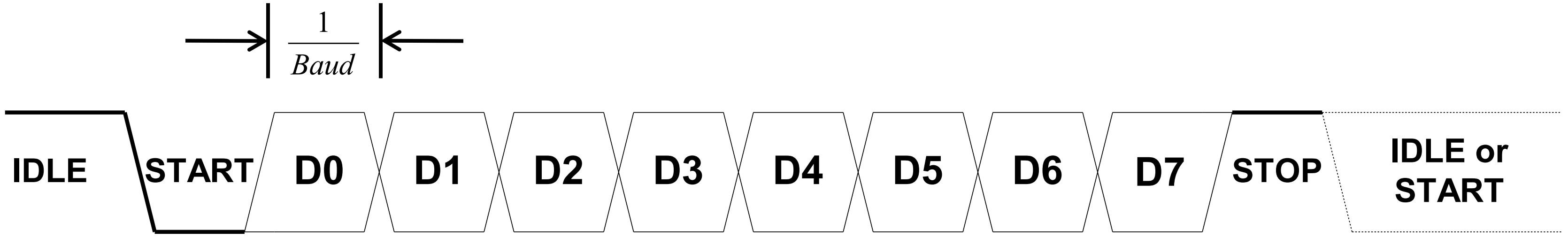
UART Interface

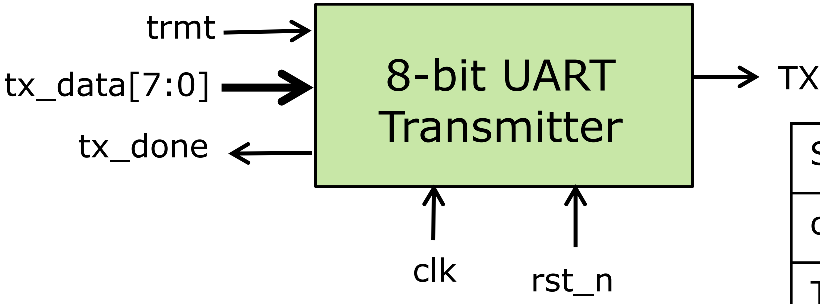
- Universal Asynchronous Receiver / Transmitter
  - Serial communication protocol
  - Agreed upon **Baud rate** & **frame format** between TX and RX
    - Standard baud rates (symbols/sec): 4800, 9600, **19200**, 38400, 57600, 115200, 230400, 460800, 921600
    - Frame formats: 1 start bit, **0** or 1 parity bit, **1** or 2 stop bit
      - 8N1: 1 start bit, no parity bit, 1 stop bit





Exercise 11: UART Transmitter

- Implement a UART transmitter (**UART_tx.sv**) with the following interface:
  - Baud rate: 19200
  - Frame format: 8N1 (1 start, 8 data, no parity, 1 stop)



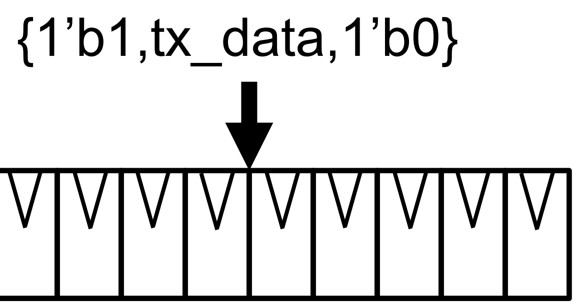
Signal	Dir	Description
clk,rst_n	in	50MHz system clock & active low reset
TX	out	Serial data output
trmt	in	Asserted for 1 clock to initiate transmission
tx_data[7:0]	in	Byte to transmit
tx_done	out	Asserted when byte is done transmitting. Stays high till next byte transmitted.

- Write a simple testbench (**UART_tx_tb.sv**) to test the functionality
  - Send a few bytes & verify baud rate and transmitted data by looking at the waveform
  - We will make a self-checking testbench next exercise.

Ex11: UART Transmitter Design Hints

Where to start?

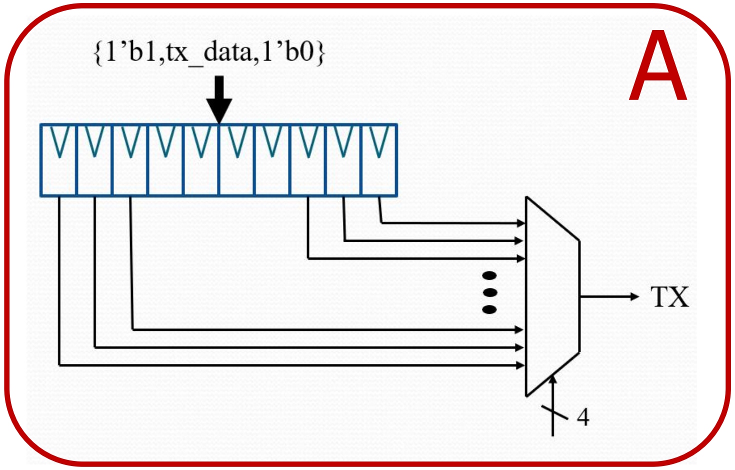
- We need to take parallel data & transmit it serially
  - Something to hold data frame → registers

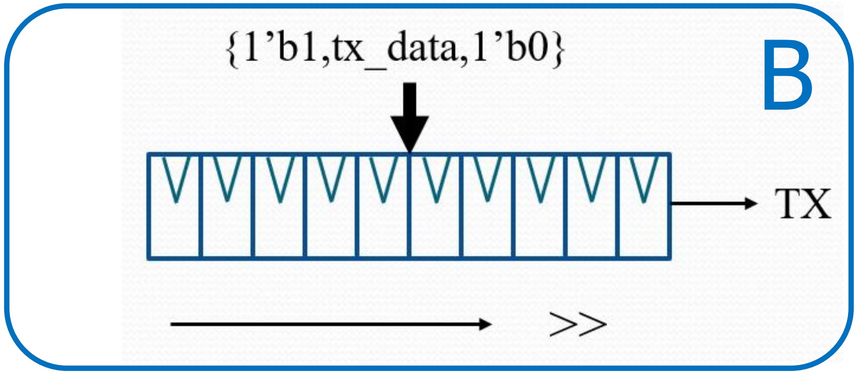


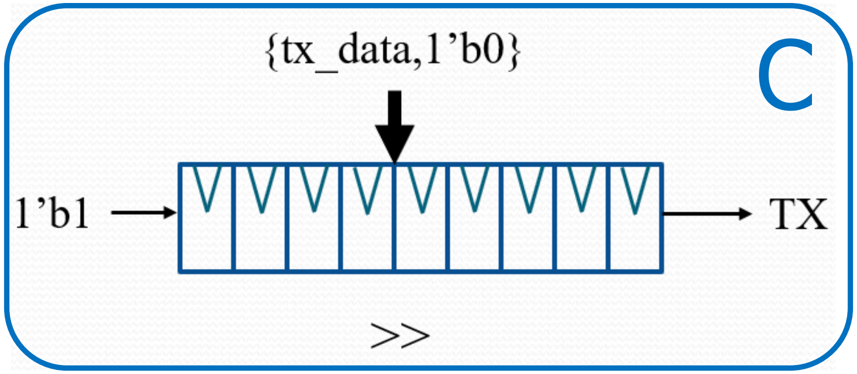
Ex11: UART Transmitter Design Hints

Where to start?

- We need to take parallel data & transmit it serially
  - Something to hold data frame → registers
  - Send 1 bit out at a time → Select or shift → Select lower area design





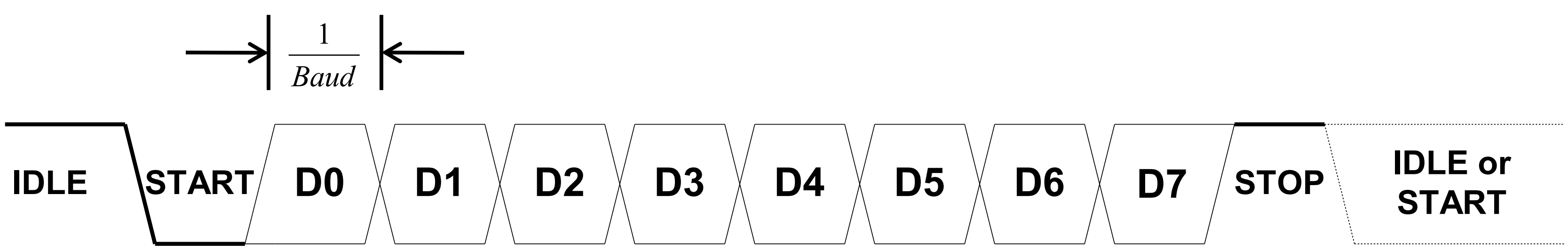


Ex11: UART Transmitter Design Hints

Where to start?

- We need to take parallel data & transmit it serially
  - Something to hold data frame → registers
  - Send 1 bit out at a time → Select or shift → Select lower area design
- We need to keep track of which bit is being transmitted
  - Counter – 10 bits in a frame
- We need a timer / clk divider for Baud rate
  - Baud rate = 19200
  - System clk = 50 MHz

→ 50MHz / 19200 = 2604 → 12-bit counter



Ex11: UART Transmitter Design Hints

Where to start?

- We need to take parallel data & transmit it serially
  - Something to hold data frame → registers
  - Send 1 bit out at a time → Select or shift → Select lower area design
- We need to keep track of which bit is being transmitted
  - Counter – 10 bits in a frame
- We need a timer / clk divider for Baud rate
  - Baud rate = 19200
  - System clk = 50 MHz

  → 50MHz / 19200 = 2604 → 12-bit counter
- We need an FSM to manage control signals
  - When to start shifting, assert tx_done, etc.

Possible Topology of UART_tx